

Why the participation share isn't one number

How much of the elite chess rating gap is explained by there being more men than women? The answer is 44% in one place and 77% in another, from identical data. This is why, in two versions: one with the mathematics and one without.

All figures: FIDE standard rating lists, May 2025 to April 2026, 12-month averaged ratings.
Active players only, no minimum rating. Comparisons are top-100 unless stated.

1 The short answer

The statistic everyone quotes is a fraction, and both halves of it depend on the question you asked:

share explained =
$$\frac{\text{gap you would expect from sample size alone}}{\text{gap you actually observe}}$$

Top: a simulation where men and women are drawn from one shared ability distribution, differing only in how many are drawn.
Bottom: the real ratings.

Change which players you count and the top moves. Change which country, or how far down the ranking you cut, and the bottom moves too. So the answer is not a number, it is a range with the question attached.

2 The answer with the mathematics

Draw n players at random from a shared ability distribution F . The expected k -th best of them lands near a specific quantile:

$$\mathbb{E}[X_{(k)}] \approx F^{-1}(1 - k/n)$$

So the predicted gap at depth k is $F^{-1}(1 - k/n_m) - F^{-1}(1 - k/n_f)$.

The quantity that drives this is not the participation ratio n_m/n_f . It is the two **tail depths**, k/n , and they move independently of one another. "The top 100" therefore means something completely different depending on the pool it is taken from:

Table 1. What a top-100 comparison actually compares, in three arenas.

ARENA	MEN	WOMEN	RATIO	TOP 100 IS THIS SHARE OF THE MEN	...AND OF THE WOMEN	SHARE EXPLAINED
World	251,137	30,420	8.3:1	0.04%	0.3%	49%
Germany	15,759	1,381	11.4:1	0.6%	7.2%	56%
England	3,176	245	13.0:1	3.1%	40.8%	77%

In England, the top 100 women are **41% of the entire pool**. The comparison is being asked to difference roughly the 97th percentile of F against the 59th, which is not an extreme-value comparison so much as the tail against the upper middle. It predicts a large gap, and the observed gap is duly explained.

Note that the ratio barely moves across those three rows, from 8.3:1 to 13.0:1, while the answer moves 28 points. Ratio is not what is doing the work. Absolute pool size is. The Gaussian case shows why: since $\mathbb{E}[\max \text{ of } n] \approx \sigma\sqrt{2 \ln n}$, the predicted gap at a fixed ratio r is

$$\sigma\sqrt{2} \left(\sqrt{\ln r n_f} - \sqrt{\ln n_f} \right)$$

Concave in log n. The same imbalance buys a much larger predicted gap in a small pool than in a large one.

3 The answer without the mathematics

Picture two lists: 3,176 English men and 245 English women. Ask each list for its top 100.

For the men that is a genuine elite, roughly the best three in every hundred. For the women, "the top 100" is **41% of everyone on the list**. You have gone past the elite, past the strong players, and well into the middle of the field simply to fill the hundred slots. The hundredth English woman is rated 1689. The hundredth English man is rated 2239.

So the comparison is the very best men against a large chunk of all the women. Of course the arithmetic answers "that was mostly sample size." Any method would. It is not measuring talent so much as measuring how short the list is.

Germany runs the identical comparison with 1,381 women, so its top 100 really is a top tenth, and the answer falls to 56%. Take it worldwide, where the top 100 women are the best 0.3% of 30,420, and it falls to 49%.

77% answers "what happens in a national scene where the women's list is short?" 49% answers "what happens at the global elite?" Both are true. They are answers to different questions, and the difference between them is not a disagreement about chess.

4 So where does it actually move, and where doesn't it

Two things move the number a great deal. How far down the ranking you cut:

Table 2. England at every depth. 3,176 men, 245 women.

CUT	TOP K IS THIS SHARE OF THE WOMEN	OBSERVED GAP	SHARE EXPLAINED	95% INTERVAL
Top 1	0.4%	342 Elo	45%	[-5, 94]
Top 10	4.1%	369 Elo	71%	[38, 104]
Top 25	10.2%	409 Elo	77%	[55, 98]
Top 50	20.4%	444 Elo	79%	[64, 93]
Top 100	40.8%	493 Elo	77%	[66, 87]

And which country you look at:

Table 3. One identical method, nine federations, top-100.

FEDERATION	MEN	WOMEN	RATIO	SHARE EXPLAINED	95% INTERVAL
Russia	8,358	2,138	3.9:1	44%	[33, 56]
Germany	15,759	1,381	11.4:1	56%	[47, 65]
Poland	7,129	1,065	6.7:1	69%	[56, 81]
Spain	20,812	1,190	17.5:1	69%	[60, 78]
Italy	8,466	495	17.1:1	72%	[63, 81]
France	19,596	1,708	11.5:1	72%	[61, 83]
England	3,176	245	13.0:1	77%	[66, 87]
India	22,402	3,508	6.4:1	77%	[62, 93]
United States	6,253	572	10.9:1	79%	[68, 90]

Range 44% to 79%, median 72%, against 49% for the world pool at the same depth. Every one of those is the same simulation on the same data. Reading the table as a ranking of national chess cultures would be a mistake: Russia sits lowest because it has the deepest women's pool relative to its men's, not because Russian women are stronger.

What does *not* move it much, and is worth saying because it is the usual objection: the shape of the assumed distribution. Swapping the real rating distribution for a fitted normal moves the world figure by about three points. The size and definition of the pool matters roughly seven times more.

5 What these numbers do and don't cover

Everything here counts players who hold a **FIDE standard rating** and are flagged as currently competing. That is a much narrower group than "people who play chess," in three ways worth stating before anyone quotes 245 back as the number of women playing chess in England.

FIDE publishes rapid and blitz as separate rating lists, and those are not included here, so a player rated only at faster time controls does not appear at all. That exclusion falls hardest on juniors and on recently registered players. FIDE registration also requires entry into FIDE-rated events, so national federation lists are considerably larger than the FIDE list for the same country. And 245 is the currently-active count: 440 English women hold a FIDE standard rating in total.

SUMMARY

The share of the elite chess gap attributable to participation is not a constant. On this data it runs from roughly **45% to 79%** depending on the arena and the depth of the cut, and the choice that moves it most is not a statistical assumption but a definitional one: **who counts as a player**.

A single quoted percentage is a function collapsed into a point. Ask which population it was computed over, and how far down the ranking it cut, before comparing it to any other.

METHOD AND DATA

FIDE standard rating lists, monthly snapshots May 2025 to April 2026, per-player mean rating across the twelve months. Active players only, as flagged by FIDE. No minimum rating applied.

Predicted gaps come from an order-statistic Monte Carlo: 6,000 simulations drawing from the pooled empirical rating distribution of both sexes, with sample sizes equal to the real male and female pools. Intervals are 95% Monte Carlo. Full code and data pipeline in the project repository.